DAC714



SBAS032A – JULY 1997 – REVISED NOVEMBER 2005

16-Bit DIGITAL-TO-ANALOG CONVERTER With Serial Data Interface

FEATURES:

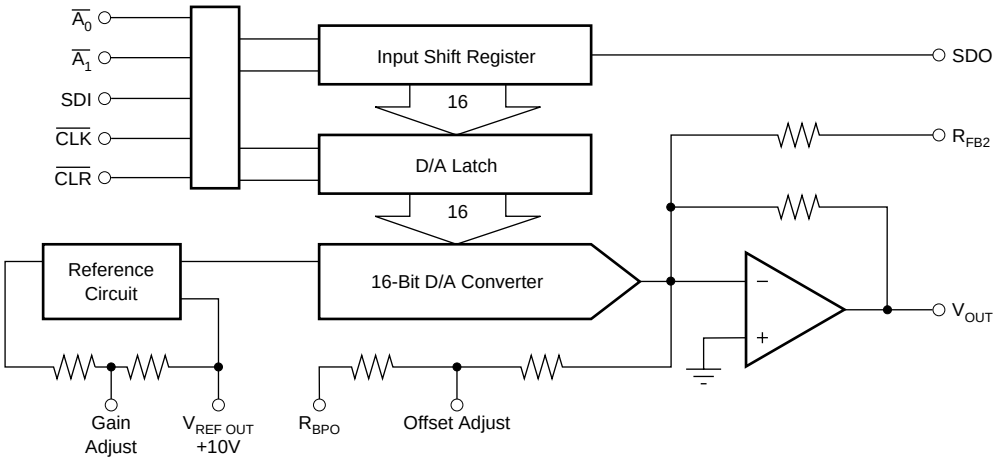
- **SERIAL DIGITAL INTERFACE**
- **VOLTAGE OUTPUT: ±10V, ±5V, 0 to +10V**
- **±1 LSB INTEGRAL LINEARITY**
- **16-BIT MONOTONIC OVER TEMPERATURE**
- **PRECISION INTERNAL REFERENCE**
- **LOW NOISE: 120nV/√Hz Including Reference**
- **16-LEAD PLASTIC AND CERAMIC SKINNY DIP AND PLASTIC SO PACKAGES**

DESCRIPTION

The DAC714 is a complete monolithic digital-to-analog (D/A) converter including a +10V temperature compensated reference, current-to-voltage amplifier, a high-speed synchronous serial interface, a serial output which allows cascading multiple converters, and an asynchronous clear function which immediately sets the output voltage to midscale.

The output voltage range is ±10V, ±5V, or 0 to +10V while operating from ±12V or ±15V supplies. The gain and bipolar offset adjustments are designed so that they can be set via external potentiometers or external D/A converters. The output amplifier is protected against short circuit to ground.

The 16-pin DAC714 is available in a plastic 0.3" DIP, ceramic 0.3" CERDIP, and wide-body plastic SO package. The DAC714P, U, HB, and HC are specified over the –40°C to +85°C temperature range while the DAC714HL is specified over the 0°C to +70°C range.



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ABSOLUTE MAXIMUM RATINGS(1)

$+V_{CC}$ to Common	0V to +17V
$-V_{CC}$ to Common	0V to –17V
$+V_{CC}$ to $-V_{CC}$	34V
ACOM to DCOM	±0.5V
Digital Inputs to Common	–1V to (V_{CC} –0.7V)
External Voltage Applied to BPO and Range Resistors	$\pm V_{CC}$
$V_{REF\ OUT}$	Indefinite Short to Common
V_{OUT}	Indefinite Short to Common
SDO	Indefinite Short to Common
Power Dissipation	750mW
Storage Temperature	–60°C to +150°C
Lead Temperature (soldering, 10s)	+300°C

NOTE: (1) Stresses above those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. Exposure to absolute maximum conditions for extended periods may affect device reliability.

ELECTROSTATIC DISCHARGE SENSITIVITY

This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

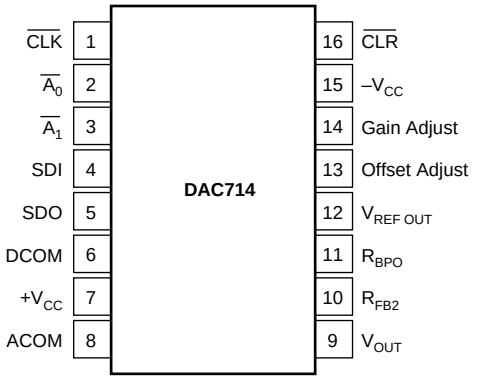
ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

PACKAGE/ORDERING INFORMATION

For the most current package and ordering information, see the Package Option Addendum at the end of this document, or see the TI website at www.ti.com.

PIN CONFIGURATION

Top View **SO/DIP**



PIN DESCRIPTIONS

PIN	LABEL	DESCRIPTION
1	$\overline{CLK}$	Serial Data Clock
2	$\overline{A_0}$	Enable for Input Register (Active Low)
3	$\overline{A_1}$	Enable for D/A Latch (Active Low)
4	SDI	Serial Data Input
5	SDO	Serial Data Output
6	DCOM	Digital Ground
7	$+V_{CC}$	Positive Power Supply
8	ACOM	Analog Ground
9	V_{OUT}	D/A Output
10	R_{FB2}	±10V Range Feedback Output
11	R_{BPO}	Bipolar Offset
12	$V_{REF\ OUT}$	Voltage Reference Output
13	Offset Adjust	Offset Adjust
14	Gain Adjust	Gain Adjust
15	$-V_{CC}$	Negative Power Supply
16	$\overline{CLR}$	Clear

ELECTRICAL CHARACTERISTICS

At T_A = +25°C, $+V_{CC}$ = +12V and +15V, $-V_{CC}$ = –12V, and –15V, unless otherwise noted.

PARAMETER	DAC714P, U			DAC714HB			DAC714HC			DAC714HL			UNITS
	MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
TRANSFER CHARACTERISTICS													
ACCURACY													
Linearity Error			±4			±2			±1			±1	LSB
T_{MIN} to T_{MAX}			±8			±4			±2			±2	LSB
Differential Linearity Error			±4			±2			±1			±1	LSB
T_{MIN} to T_{MAX}			±8			±4			±2			±1	LSB
Monotonicity	14			15			16			16			Bits
Monotonicity Over Spec Temp Range	13			14			15			16			Bits
Gain Error[3]			±0.1			±0.1			±0.1			±0.1	%
T_{MIN} to T_{MAX}			±0.25			±0.25			±0.25			±0.25	%
Unipolar/Bipolar Zero Error[3]			±0.1			±0.1			±0.1			±0.1	% of FSR[2]
T_{MIN} to T_{MAX}			±0.2			±0.2			±0.2			±0.2	% of FSR
Power Supply Sensitivity of Gain			±0.003			±0.003			±0.003			±0.003	%FSR/%V_{CC}
			±30			±30			±30			±30	ppm FSR/%V_{CC}
DYNAMIC PERFORMANCE													
Settling Time													
(to ±0.003%FSR, 5kΩ || 500pF Load)[4]													
20V Output Step		6	10		6	10		6	10		6	10	µs
1LSB Output Step[5]		4			4			4			4		µs
Output Slew Rate		10			10			10			10		V/µs
Total Harmonic Distortion													
0dB, 1001Hz, f_S = 100kHz		0.005			0.005			0.005			0.005		%
–20dB, 1001Hz, f_S = 100kHz		0.03			0.03			0.03			0.03		%
–60dB, 1001Hz, f_S = 100kHz		3.0			3.0			3.0			3.0		%
SINAD: 1001Hz, f_S = 100kHz		87			87			87			87		dB
Digital Feedthrough[5]		2			2			2			2		nV–s
Digital-to-Analog Glitch Impulse[5]		15			15			15			15		nV–s
Output Noise Voltage (includes reference)		120			120			120			120		nV/√Hz
ANALOG OUTPUT													
Output Voltage Range													
$+V_{CC}$, $-V_{CC}$ = ±11.4V	±10			±10			±10			±10			V
Output Current	±5			±5			±5			±5			mA
Output Impedance		0.1			0.1			0.1			0.1		Ω
Short Circuit to ACOM Duration		Indefinite			Indefinite			Indefinite			Indefinite		
REFERENCE VOLTAGE													
Voltage	+9.975	+10.000	+10.025	+9.975	+10.000	+10.025	+9.975	+10.000	+10.025	+9.975	+10.000	+10.025	V
T_{MIN} to T_{MAX}	+9.960		+10.040	+9.960		+10.040	+9.960		+10.040	+9.960		+10.040	V
Output Resistance		1			1			1			1		Ω
Source Current	2			2			2			2			mA
Short Circuit to ACOM Duration		Indefinite			Indefinite			Indefinite			Indefinite		
INTERFACE													
RESOLUTION		16			16			16			16		Bits
DIGITAL INPUTS													
Serial Data Input Code													
Logic Levels[1]						Binary Two's Complement							
V_{IH}	+2.0		(V_{CC} –1.4)	+2.0		(V_{CC} –1.4)	+2.0		(V_{CC} –1.4)	+2.0		(V_{CC} –1.4)	V
V_{IL}	0		+0.8	0		+0.8	0		+0.8	0		+0.8	V
I_{IH} (V_I = +2.7V)			±10			±10			±10			±10	µA
I_{IL} (V_I = +0.4V)			±10			±10			±10			±10	µA
DIGITAL OUTPUT													
Serial Data													
V_{OL} (I_{SINK} = 1.6mA)	0		+0.4	0		+0.4	0		+0.4	0		+0.4	V
V_{OH} (I_{SOURCE} = 500µA), T_{MIN} to T_{MAX}	+2.4		+5	+2.4		+5	+2.4		+5	+2.4		+5	V
POWER SUPPLY REQUIREMENTS													
Voltage													
$+V_{CC}$	+11.4	+15	+16.5	+11.4	+15	+16.5	+11.4	+15	+16.5	+11.4	+15	+16.5	V
$-V_{CC}$	–11.4	–15	–16.5	–11.4	–15	–16.5	–11.4	–15	–16.5	–11.4	–15	–16.5	V
Current (No Load, ±15V Supplies)[6]													
$+V_{CC}$		13	16		13	16		13	16		13	16	mA
$-V_{CC}$		22	26		22	26		22	26		22	26	mA
Power Dissipation[7]			625			625			625			625	mW
TEMPERATURE RANGES													
Specification													
All Grades	–40		+85	–40		+85	–40		+85	0		+70	°C
Storage	–60		+150	–60		+150	–60		+150	–60		+150	°C
Thermal Coefficient, θ_{JA}		75			75			75			75		°C/W

NOTES: (1) Digital inputs are TTL and +5V CMOS compatible over the specification temperature range. (2) FSR means Full Scale Range. For example, for ±10V output, FSR = 20V. (3) Errors externally adjustable to zero. (4) Maximum represents the 3σ limit. Not 100% tested for this parameter. (5) For the worst-case Binary Two's Complement code changes: $FFFF_H$ to 0000_H and 0000_H to $FFFF_H$. (6) During power supply turn on, the transient supply current may approach 3x the maximum quiescent specification. (7) Typical (i.e. rated) supply voltages times maximum currents.

TIMING SPECIFICATIONS

T_A = –40°C to +85°C, $+V_{CC}$ = +12V or +15V, $–V_{CC}$ = –12V or –15V.

SYMBOL	PARAMETER	MIN	MAX	UNITS
t_{CLK}	Data Clock Period	100		ns
t_{CL}	Clock LOW	50		ns
t_{CH}	Clock HIGH	50		ns
t_{A0S}	Setup Time for $\overline{A_0}$	50		ns
t_{A1S}	Setup Time for $\overline{A_1}$	50		ns
t_{AOH}	Hold Time for $\overline{A_0}$	0		ns
t_{A1H}	Hold Time for $\overline{A_1}$	0		ns
t_{DS}	Setup Time for DATA	50		ns
t_{DH}	Hold Time for DATA	10		ns
t_{DSOP}	Output Propagation Delay		140	ns
t_{CP}	Clear Pulsewidth	200		ns

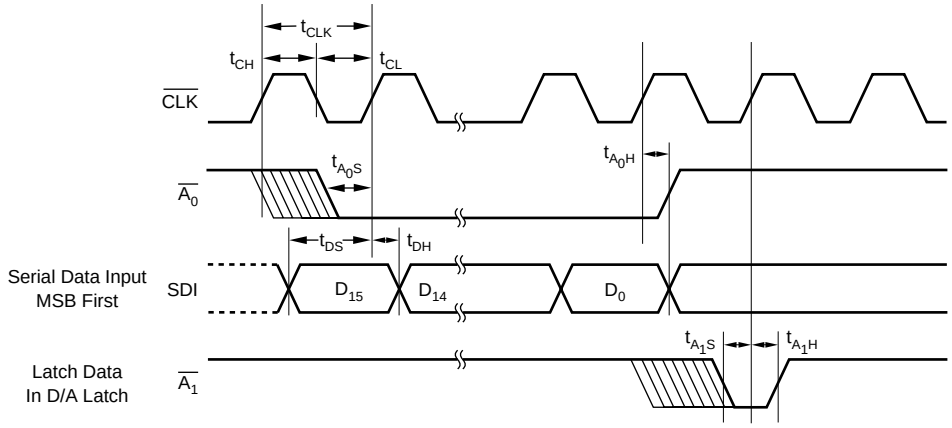
TRUTH TABLE

$\overline{A_0}$	$\overline{A_1}$	$\overline{CLK}$	$\overline{CLR}$	DESCRIPTION
0	1	1 → 0 → 1	1	Shift Serial Data into SDI
1	0	1 → 0 → 1	1	Load D/A Latch
1	1	1 → 0 → 1	1	No Change
0	0	1 → 0 → 1	1	Two Wire Operation(1)
X	X	1	1	No Change
X	X	X	0	Reset D/A Latch

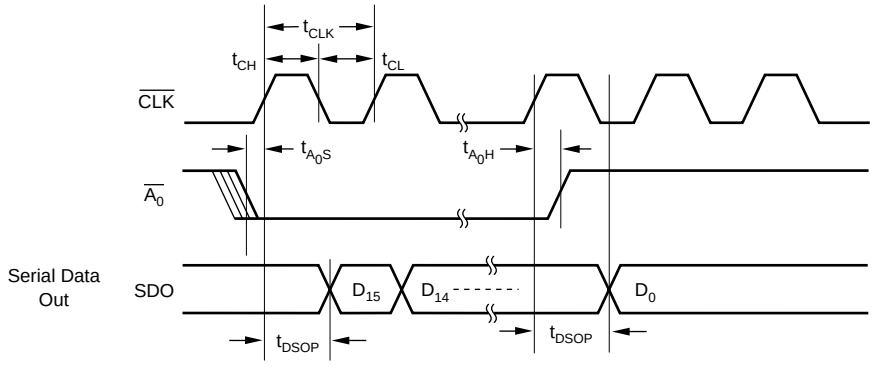
NOTES: X = Don't Care. (1) All digital input changes will appear at the output.

TIMING DIAGRAMS

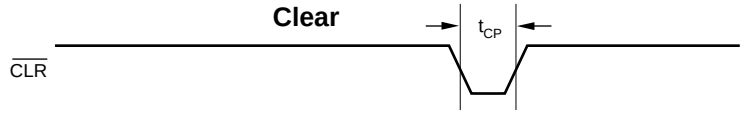
Serial Data In



Serial Data Out



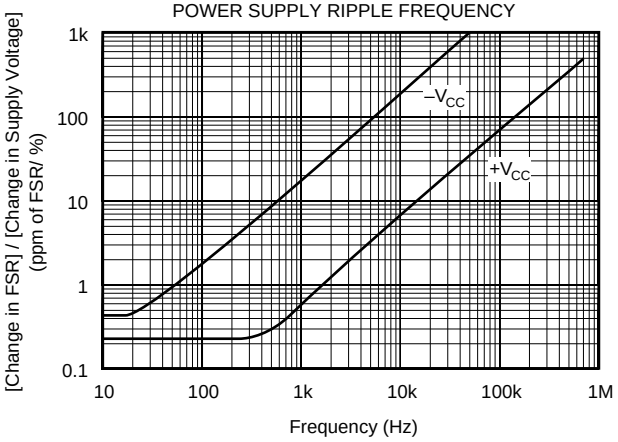
Clear



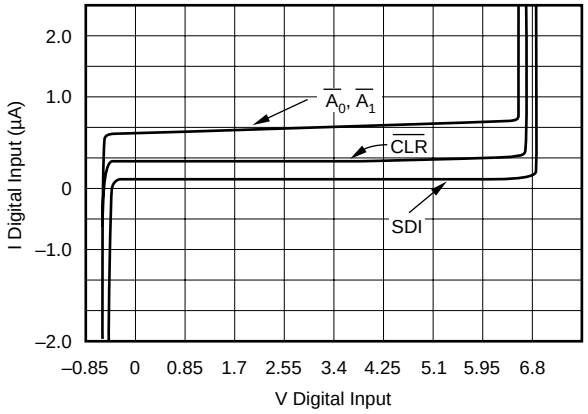
TYPICAL CHARACTERISTICS

At T_A = +25°C, V_{CC} = ±15V, unless otherwise noted.

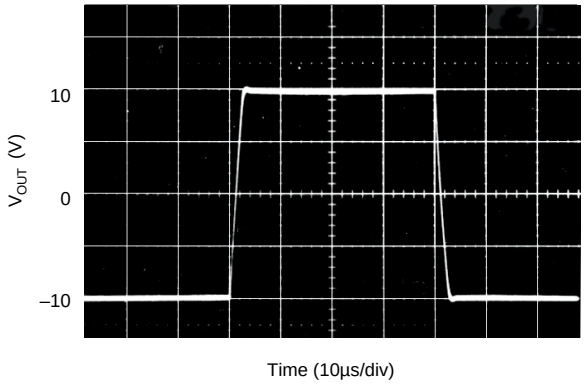
POWER SUPPLY REJECTION vs
POWER SUPPLY RIPPLE FREQUENCY



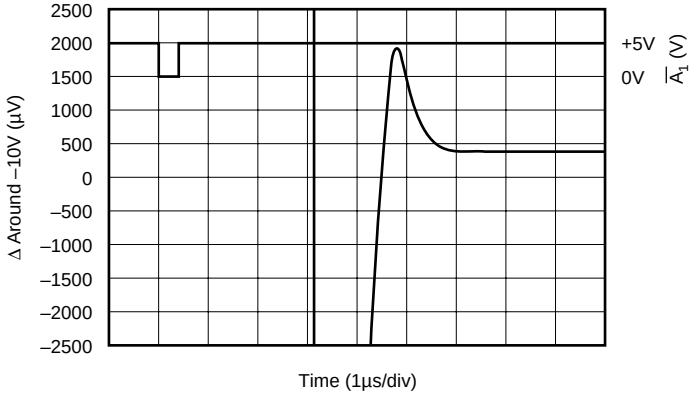
LOGIC vs V LEVEL



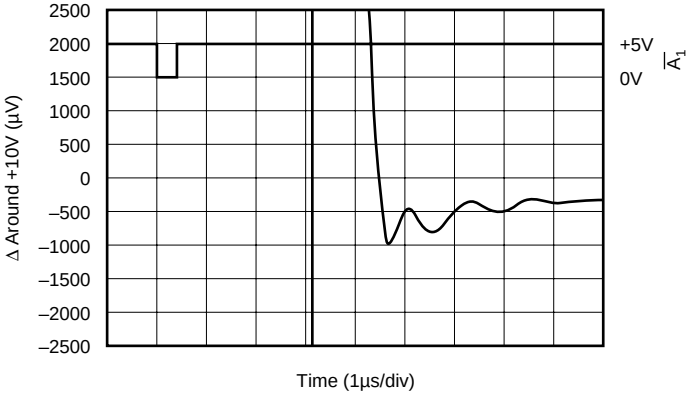
± FULL SCALE OUTPUT SWING



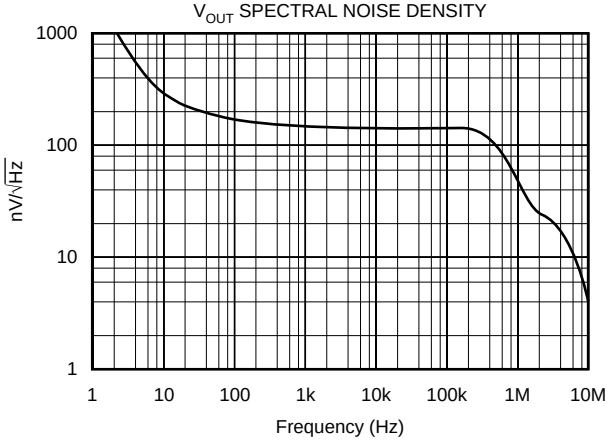
SETTLING TIME, +10V TO –10V



SETTLING TIME, –10V TO +10V



VOUT SPECTRAL NOISE DENSITY



DISCUSSION OF SPECIFICATIONS

LINEARITY ERROR

Linearity error is defined as the deviation of the analog output from a straight line drawn between the end points of the transfer characteristic.

DIFFERENTIAL LINEARITY ERROR

Differential linearity error (DLE) is the deviation from 1LSB of an output change from one adjacent state to the next. A DLE specification of ±1/2LSB means that the output step size can range from 1/2LSB to 3/2LSB when the digital input code changes from one code word to the adjacent code word. If the DLE is more positive than –1LSB, the D/A is said to be monotonic.

MONOTONICITY

A D/A converter is monotonic if the output either increases or remains the same for increasing digital input values. Monotonicity of the C and L grades is assured over the specification temperature range to 16 bits.

SETTLING TIME

Settling time is the total time (including slew time) for the D/A output to settle to within an error band around its final value after a change in input. Settling times are specified to within ±0.003% of Full Scale Range (FSR) for an output step change of 20V and 1LSB. The 1LSB change is measured at the Major Carry ($FFFF_H$ to 0000_H, and 0000_H to $FFFF_H$: BTC codes), the input transition at which worst-case settling time occurs.

TOTAL HARMONIC DISTORTION

Total harmonic distortion is defined as the ratio of the square root of the sum of the squares of the values of the harmonics to the value of the fundamental frequency. It is expressed in % of the fundamental frequency amplitude at sampling rate f_S.

SIGNAL-TO-NOISE AND DISTORTION RATIO (SINAD)

SINAD includes all the harmonic and outstanding spurious components in the definition of output noise power in addition to quantizing and internal random noise power. SINAD is expressed in dB at a specified input frequency and sampling rate, f_S.

DIGITAL-TO-ANALOG GLITCH IMPULSE

The amount of charge injected into the analog output from the digital inputs when the inputs change state. It is measured at half scale at the input codes where as many as possible switches change state—from 0000_H to $FFFF_H$.

DIGITAL FEEDTHROUGH

When the A/D is not selected, high frequency logic activity on the digital inputs is coupled through the device and shows up as output noise. This noise is digital feedthrough.

OPERATION

The DAC714 is a monolithic integrated-circuit 16-bit D/A converter complete with 16-bit D/A switches and ladder network, voltage reference, output amplifier and a serial interface.

INTERFACE LOGIC

The DAC714 has double-buffered data latches. The input data latch holds a 16-bit data word before loading it into the second latch, the D/A latch. This double-buffered organization permits simultaneous update of several D/A converters. All digital control inputs are active low. Refer to the block diagram shown in Figure 1.

All latches are level-triggered. Data present when the enable inputs are logic “0” will enter the latch. When the enable inputs return to logic “1”, the data is latched.

The $\overline{CLR}$ input resets both the input latch and the D/A latch to 0000_H (midscale).

LOGIC INPUT COMPATIBILITY

The DAC714 digital inputs are TTL compatible (1.4V switching level), low leakage, and high impedance. Thus, the inputs are suitable for being driven by any type of 5V logic family, such as CMOS. An equivalent circuit for the digital inputs is shown in Figure 2.

The inputs will float to logic “0” if left unconnected. It is recommended that any unused inputs be connected to DCOM to improve noise immunity.

Digital inputs remain high impedance when power is off.

INPUT CODING

The DAC714 is designed to accept binary two’s complement (BTC) input codes with the MSB first which are compatible with bipolar analog output operation. For this configuration, a digital input of $7FFF_H$ produces a plus full scale output, 8000_H produces a minus full scale output, and 0000_H produces bipolar zero output.

INTERNAL REFERENCE

The DAC714 contains a +10V reference. The reference output may be used to drive external loads, sourcing up to 2mA. The load current should be constant; otherwise, the gain and bipolar offset of the converter will vary.

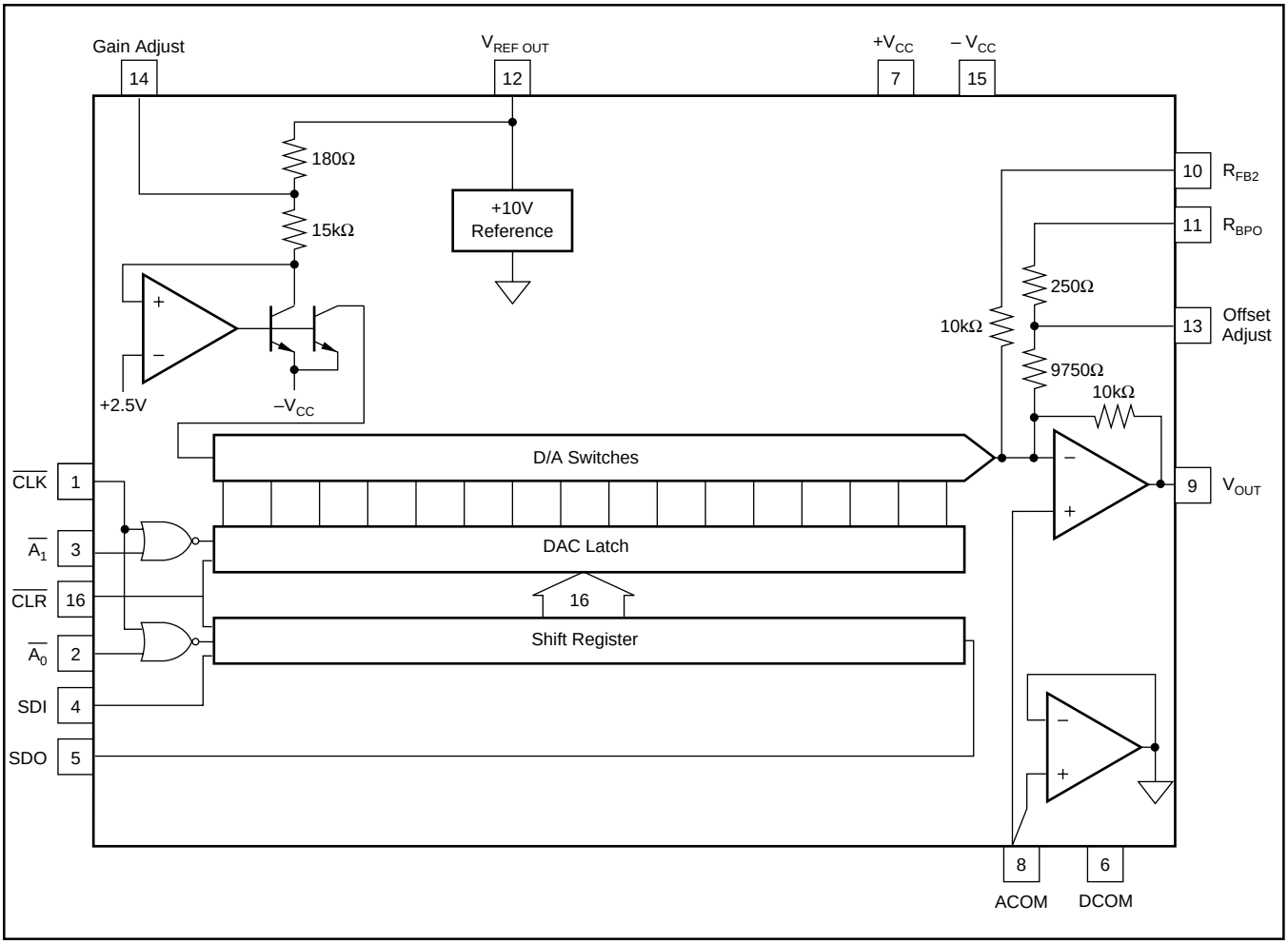


FIGURE 1. DAC714 Block Diagram.

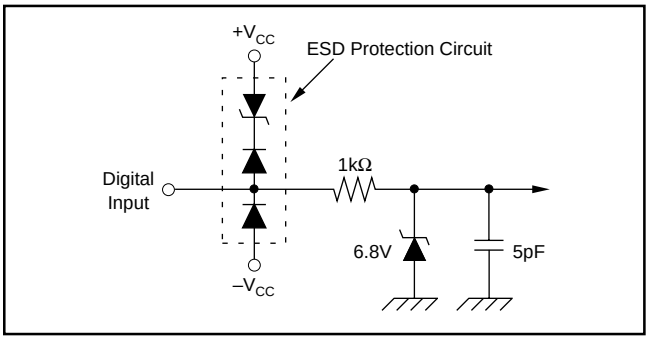


FIGURE 2. Equivalent Circuit of Digital Inputs.

OUTPUT VOLTAGE SWING

The output amplifier of the DAC714 is designed to achieve a ±10V output range while operating on ±11.4V or higher power supplies.

GAIN AND OFFSET ADJUSTMENTS

Figure 3 illustrates the relationship of offset and gain adjustments for a bipolar connected D/A converter. Offset should be adjusted first to avoid interaction of adjustments. See Table I for calibration values and codes. These adjustments have a minimum range of ±0.3%.

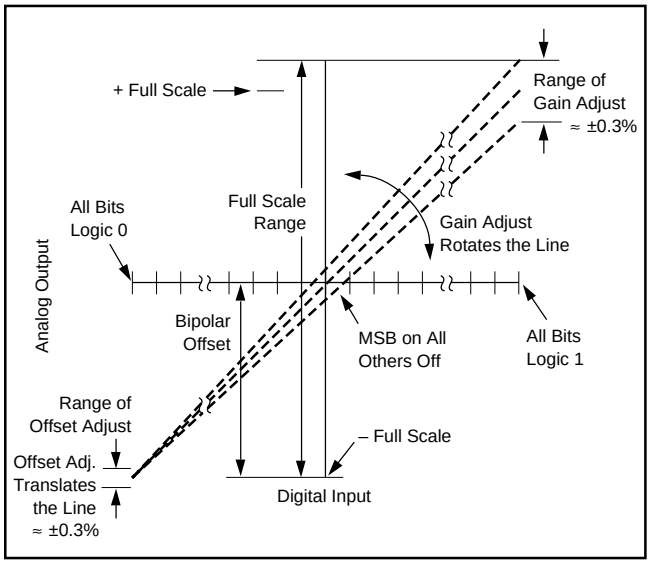


FIGURE 3. Relationship of Offset and Gain Adjustments.

Offset Adjustment

Apply the digital input code, 8000_H, that produces the maximum negative output voltage and adjust the offset potentiometer or the offset adjust D/A converter for –10V (or 0V unipolar).

DAC714 CALIBRATION VALUES			
DIGITAL INPUT CODE BINARY TWO'S COMPLEMENT, BTC	ANALOG OUTPUT (V) BIPOLAR 20V RANGE	ANALOG OUTPUT (V) UNIPOLAR 10V RANGE	DESCRIPTION
$7FFF_H$	+9.999695	+9.999847	+ Full Scale –1LSB
4000_H	+5.000000	+7.500000	3/4 Scale
0001_H	+0.000305	+5.000153	BPZ + 1LSB
0000_H	0.000000	+5.000000	Bipolar Zero (BPZ)
$FFFF_H$	–0.000305	+4.999847	BPZ – 1LSB
$C000_H$	–5.000000	+2.500000	1/4 Scale
8000_H	–10.00000	0.000000	Minus Full Scale

TABLE I. Digital Input and Analog Output Voltage Calibration Values.

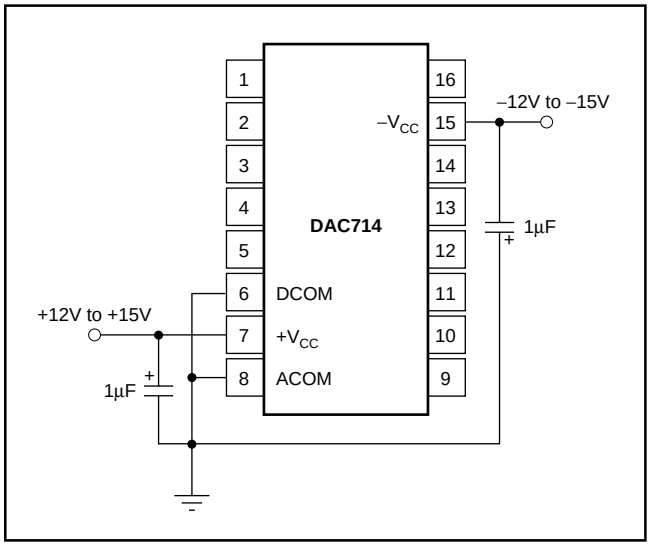


FIGURE 4. Power Supply Connections.

Gain Adjustment

Apply the digital input that gives the maximum positive voltage output. Adjust the gain potentiometer or the gain adjust D/A converter for this positive full scale voltage.

INSTALLATION

GENERAL CONSIDERATIONS

Due to the high accuracy of the DAC714 system design, problems such as grounding and contact resistance become very important. A 16-bit converter with a 20V full-scale range has a 1LSB value of 305µV. With a load current of 5mA, series wiring and connector resistance of only 60mΩ will cause a voltage drop of 300µV. To understand what this means in terms of a system layout, the resistivity of a typical 1 ounce copper-clad printed circuit board is 1/2 mΩ per square. For a 5mA load, a 10 milliinch wide printed circuit conductor 60 milliinches long will result in a voltage drop of 150µV.

The analog output of DAC714 has an LSB size of 305µV (–96dB) in the bipolar mode. The rms noise floor of the D/A should remain below this level in the frequency range of interest. The DAC714's output noise spectral density (which includes the noise contributed by the internal reference,) is shown in the Typical Characteristic section.

Wiring to high-resolution D/A converters should be routed to provide optimum isolation from sources of RFI and EMI. The key to elimination of RF radiation or pickup is small loop area. Signal leads and their return conductors should be kept close together such that they present a small capture cross-section for any external field. Wire-wrap construction is not recommended.

POWER SUPPLY AND REFERENCE CONNECTIONS

Power supply decoupling capacitors should be added as shown in Figure 4. Best performance occurs using a 1 to 10µF tantalum capacitor at $-V_{CC}$. Applications with less critical settling time may be able to use 0.01µF at $-V_{CC}$ as well as at $+V_{CC}$. The capacitors should be located close to the package.

The DAC714 has separate ANALOG COMMON and DIGITAL COMMON pins. The current through DCOM is mostly switching transients and are up to 1mA peak in amplitude. The current through ACOM is typically 5µA for all codes.

Use separate analog and digital ground planes with a single interconnection point to minimize ground loops. The analog pins are located adjacent to each other to help isolate analog from digital signals. Analog signals should be routed as far as possible from digital signals and should cross them at right angles. A solid analog ground plane around the D/A package, as well as under it in the vicinity of the analog and power supply pins, will isolate the D/A from switching currents. It is recommended that DCOM and ACOM be connected directly to the ground planes under the package.

If several DAC714s are used or if DAC714 shares supplies with other components, connecting the ACOM and DCOM lines to together once at the power supplies rather than at each chip may give better results.

LOAD CONNECTIONS

Since the reference point for V_{OUT} and $V_{REF\ OUT}$ is the ACOM pin, it is important to connect the D/A converter load directly to the ACOM pin. Refer to Figure 5.

Lead and contact resistances are represented by R_1 through R_3. As long as the load resistance R_L is constant, R_1 simply introduces a gain error and can be removed by gain adjustment of the D/A or system-wide gain calibration. R_2 is part of R_L if the output voltage is sensed at ACOM.

In some applications it is impractical to return the load to the ACOM pin of the D/A converter. Sensing the output voltage at the SYSTEM GROUND point is reasonable, because there is no change in DAC714 ACOM current, provided that R_3 is a low-resistance ground plane or conductor. In this case you may wish to connect DCOM to SYSTEM GROUND as well.

GAIN AND OFFSET ADJUST

Connections Using Potentiometers

GAIN and OFFSET adjust pins provide for trim using external potentiometers. 15-turn potentiometers provide sufficient resolution. Range of adjustment of these trims is at least ±0.3% of Full Scale Range. Refer to Figure 6.

Using D/A Converters

The GAIN ADJUST and OFFSET ADJUST circuits of the DAC714 have been arranged so that these points may be easily driven by external D/A converters. Refer to Figure 7. 12-bit D/A converters provide an OFFSET adjust resolution and a GAIN adjust resolution of 30µV to 50µV per LSB step.

Nominal values of GAIN and OFFSET occur when the D/A converters outputs are at approximately half scale, +5V.

OUTPUT VOLTAGE RANGE CONNECTIONS

The DAC714 output amplifier is connected internally to provide a 20V output range. For other ranges and configurations, see Figures 6 and 7.

DIGITAL INTERFACE

SERIAL INTERFACE

The DAC714 has a serial interface with two data buffers which can be used for either synchronous or asynchronous updating of multiple D/A converters. $\overline{A0}$ is the enable control for the input shift register. $\overline{A1}$ is the enable for the D/A Latch. $\overline{CLK}$ is used to strobe data into the latches enabled by $\overline{A0}$ and $\overline{A1}$. A $\overline{CLR}$ function is also provided and when enabled it sets the shift register and the D/A Latch to 0000_H (output voltage is midscale).

Multiple DAC714s can be connected to the same $\overline{CLK}$ and data lines in two ways. The output of the serial shift register is available as SDO so that any number of DAC714s can be cascaded on the same input bit stream as shown in Figures 8 and 9. This configuration allows all D/A converters to be updated simultaneously and requires a minimum number of control signals. These configurations do require 16N $\overline{CLK}$ cycles to load any given D/A converter, where N is the number of D/A converters.

The DAC714 can also be connected in parallel as shown in Figure 10. This configuration allows any D/A converter in the system to be updated in a maximum of 16 $\overline{CLK}$ cycles.

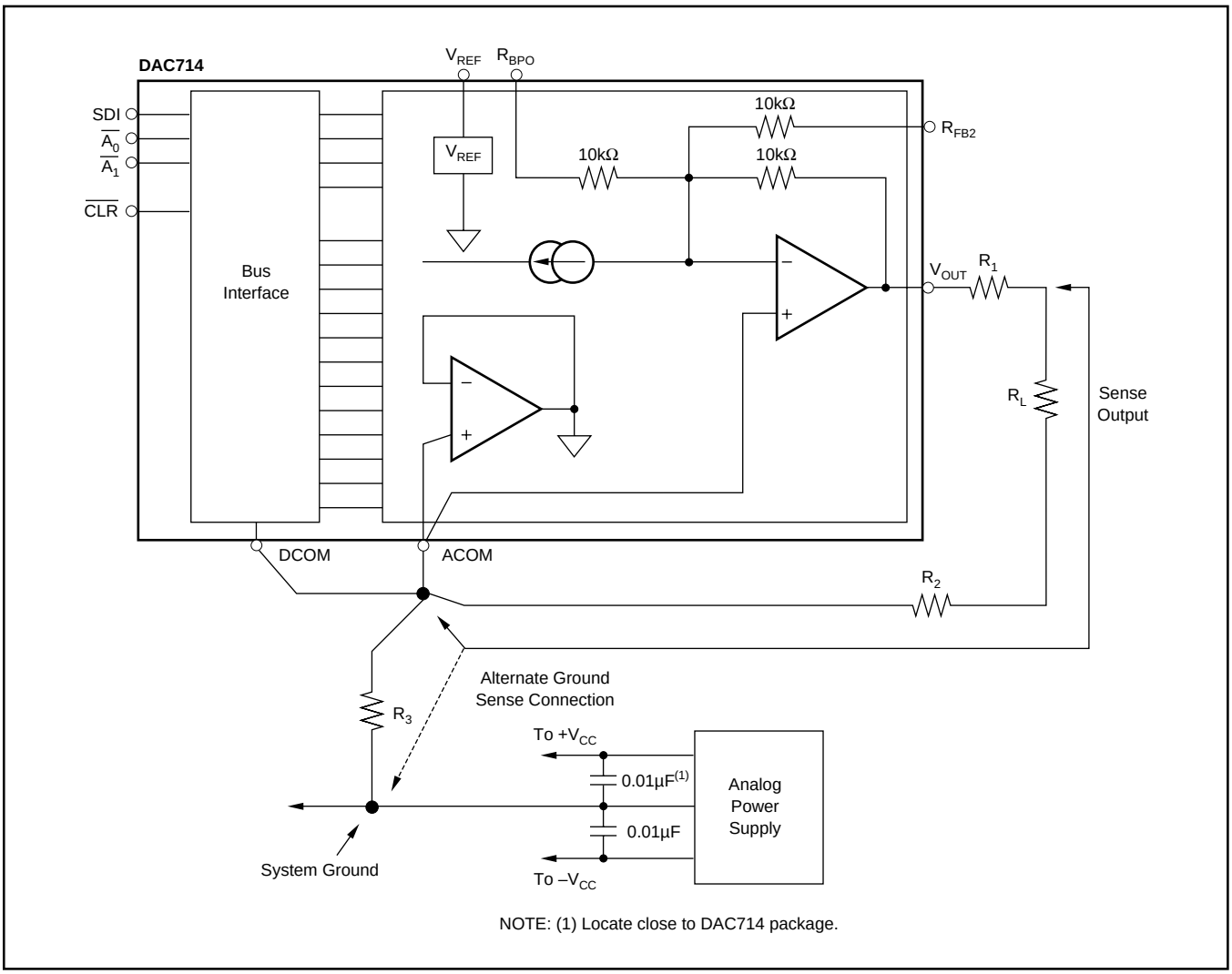
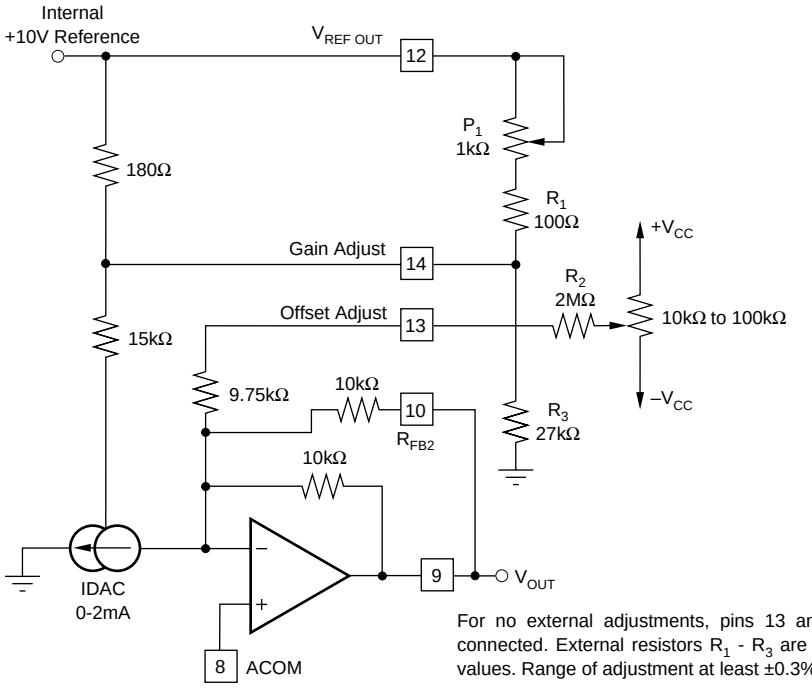
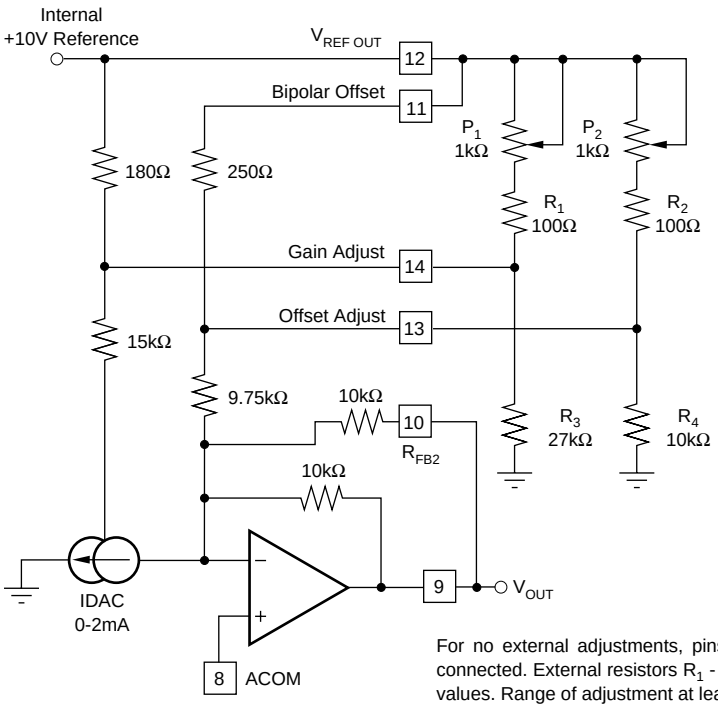


FIGURE 5. System Ground Considerations for High-Resolution D/A Converters.



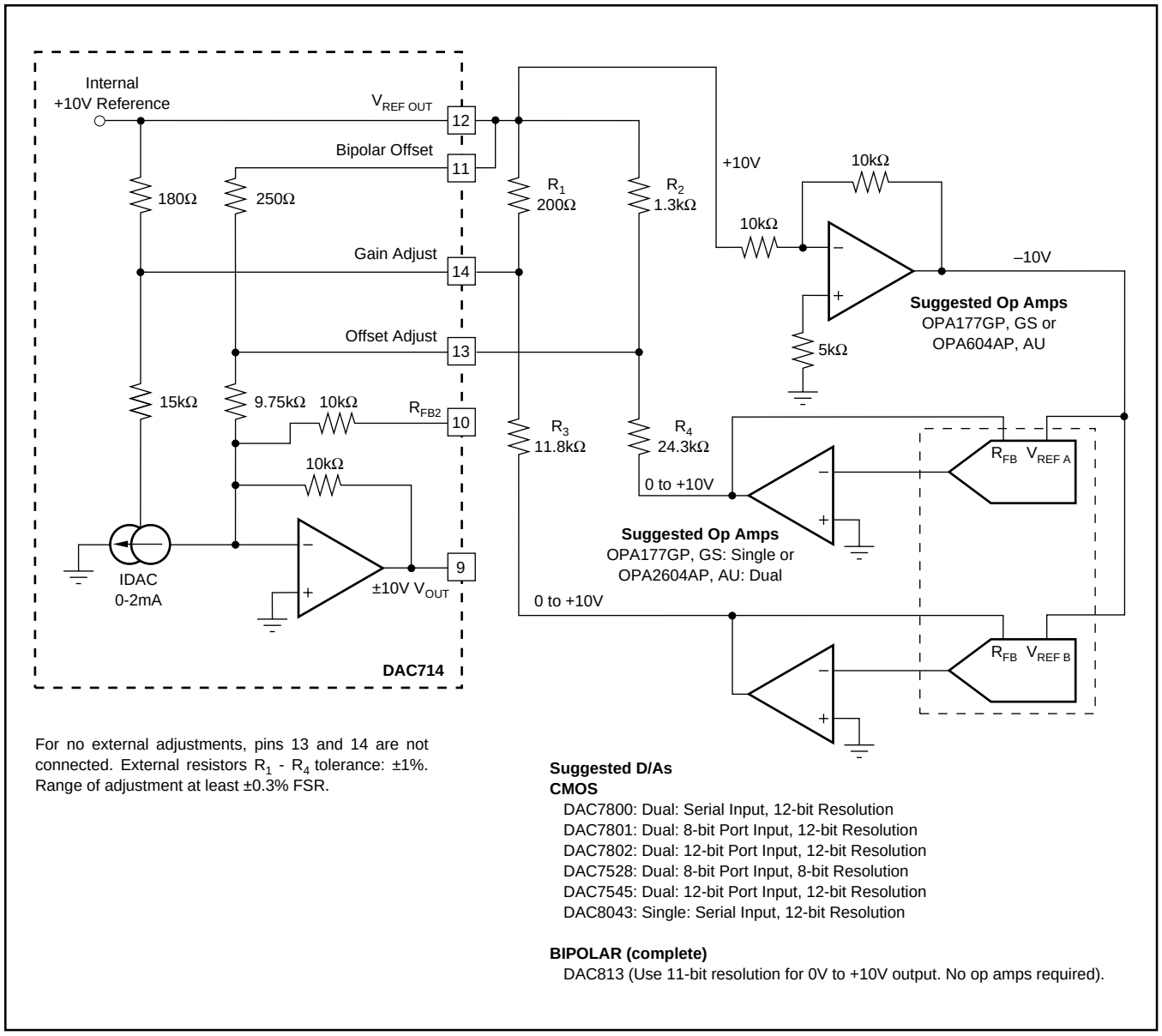
For no external adjustments, pins 13 and 14 are not connected. External resistors R_1 - R_3 are standard ±1% values. Range of adjustment at least ±0.3% FSR.

FIGURE 6a. Manual Offset and Gain Adjust Circuits; Unipolar Mode (0V to +10V output range).



For no external adjustments, pins 13 and 14 are not connected. External resistors R_1 - R_4 are standard ±1% values. Range of adjustment at least ±0.3% FSR.

FIGURE 6b. Manual Offset and Gain Adjust Circuits; Bipolar Mode (–5V to +5V output range).



For no external adjustments, pins 13 and 14 are not connected. External resistors R_1 - R_4 tolerance: ±1%. Range of adjustment at least ±0.3% FSR.

Suggested D/As

CMOS

DAC7800: Dual: Serial Input, 12-bit Resolution
DAC7801: Dual: 8-bit Port Input, 12-bit Resolution
DAC7802: Dual: 12-bit Port Input, 12-bit Resolution
DAC7528: Dual: 8-bit Port Input, 8-bit Resolution
DAC7545: Dual: 12-bit Port Input, 12-bit Resolution
DAC8043: Single: Serial Input, 12-bit Resolution

BIPOLAR (complete)

DAC813 (Use 11-bit resolution for 0V to +10V output. No op amps required).

FIGURE 7. Gain and Offset Adjustment in the Bipolar Mode Using D/A Converters (–10V to +10V output range).

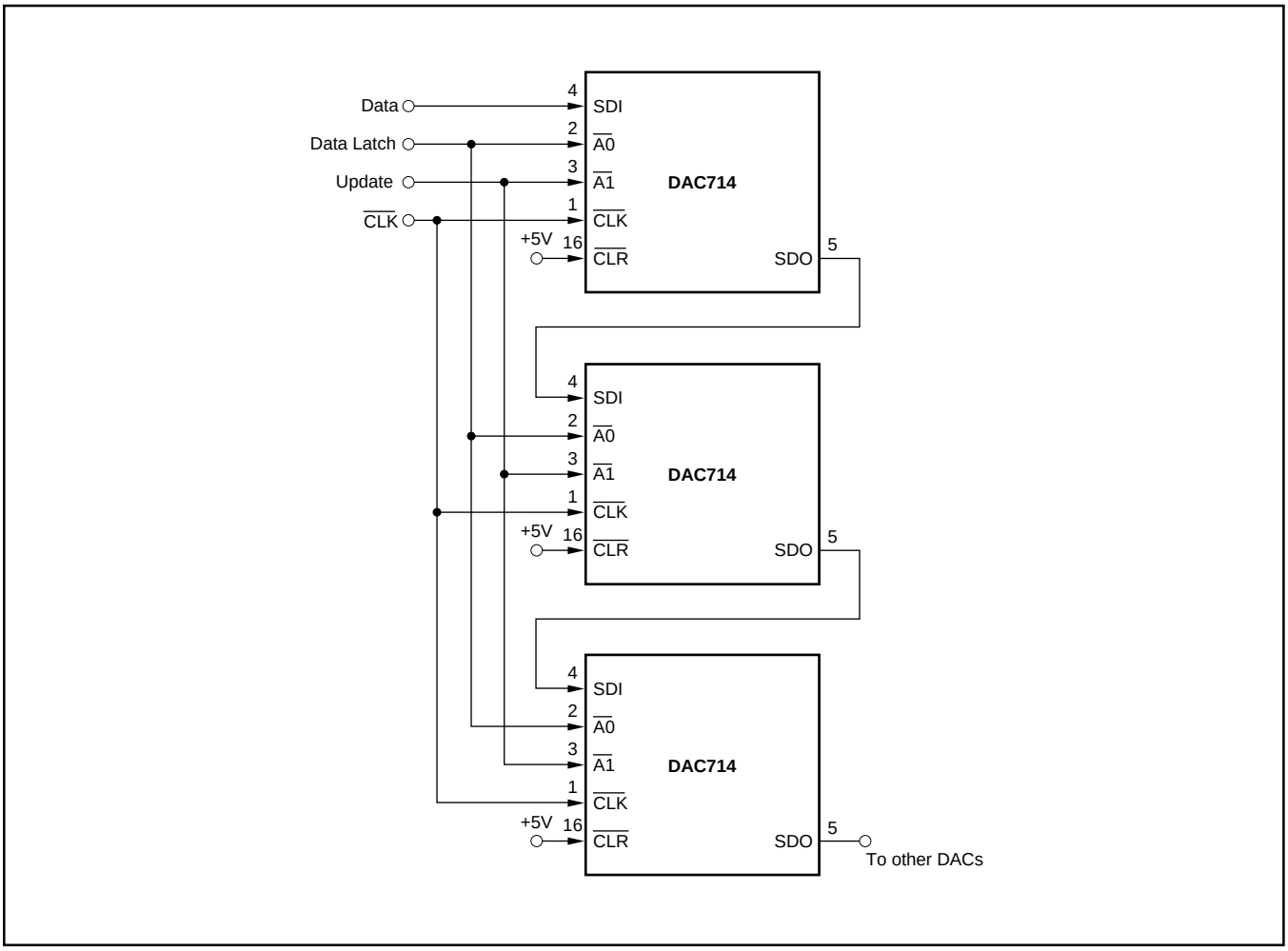


FIGURE 8a. Cascaded Serial Bus Connection with Synchronous Update.

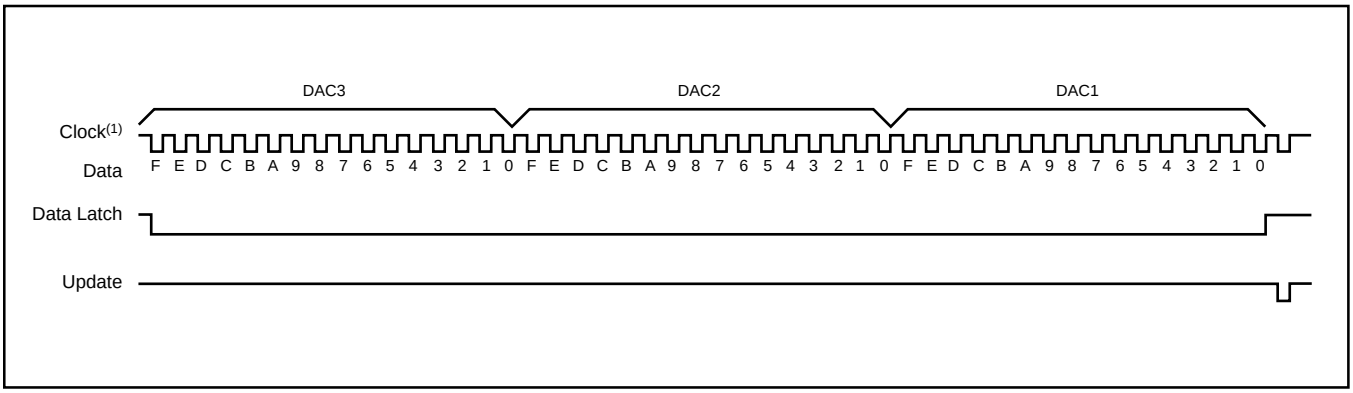


FIGURE 8b. Timing Diagram For Figure 8a.

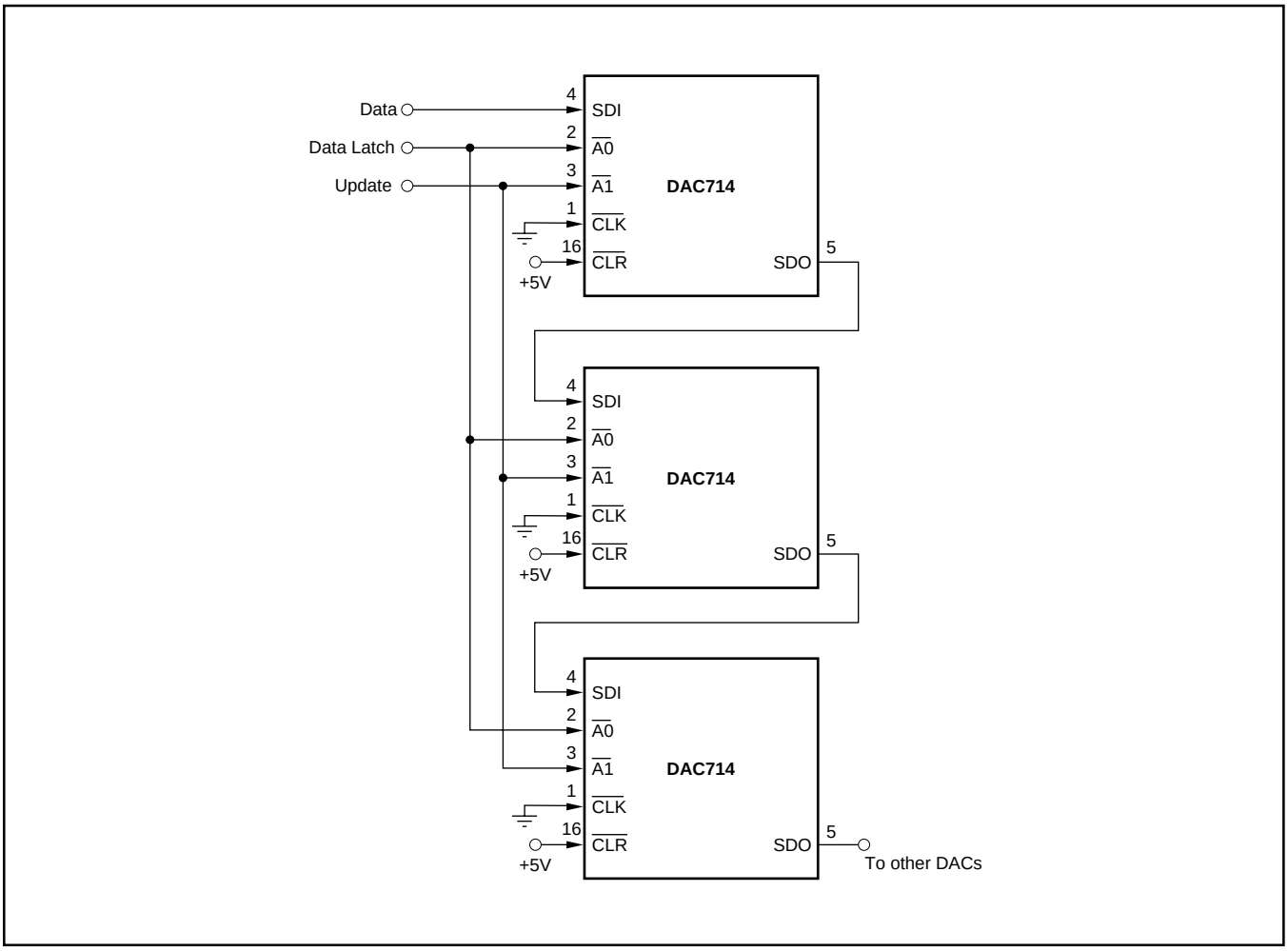


FIGURE 9a. Cascaded Serial Bus Connection with Asynchronous Update.

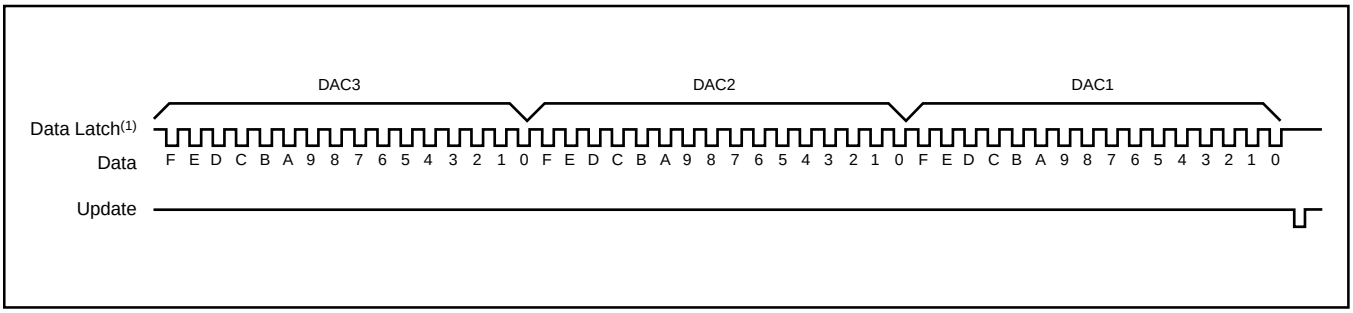


FIGURE 9b. Timing Diagram For Figure 9a.

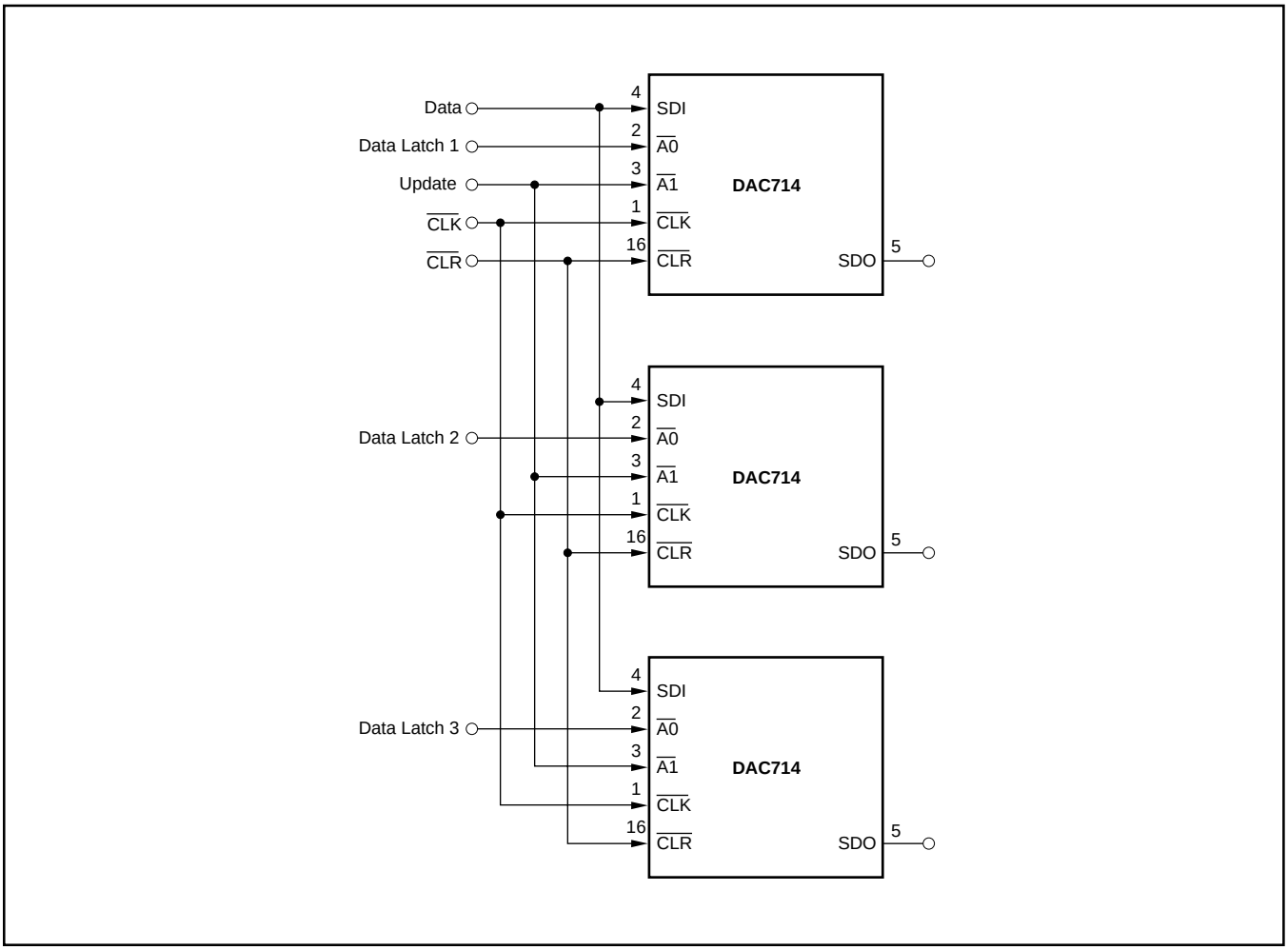


FIGURE 10a. Parallel Bus Connection.

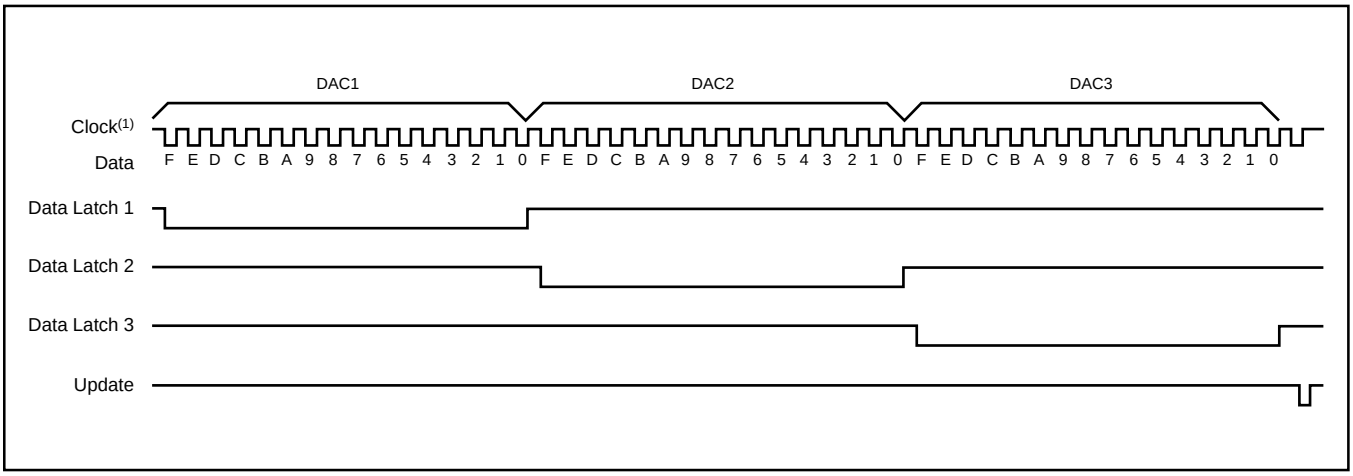


FIGURE 10b. Timing Diagram For Figure 10a.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
DAC714P	NRND	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	DAC714P
DAC714P.A	NRND	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	DAC714P
DAC714U	Active	Production	SOIC (DW) | 16	40 | TUBE	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 85	DAC714U
DAC714U.A	Active	Production	SOIC (DW) | 16	40 | TUBE	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 85	DAC714U
DAC714U/1K	Active	Production	SOIC (DW) | 16	1000 | LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 85	DAC714U
DAC714U/1K.A	Active	Production	SOIC (DW) | 16	1000 | LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 85	DAC714U
DAC714U/1KG4	Active	Production	SOIC (DW) | 16	1000 | LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 85	DAC714U

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

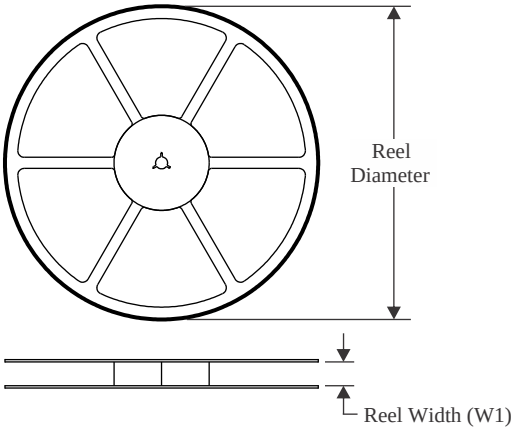
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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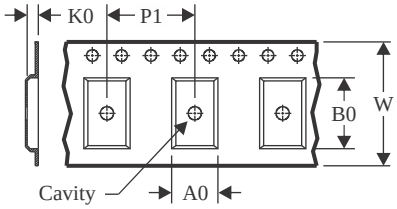
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TAPE AND REEL INFORMATION

REEL DIMENSIONS

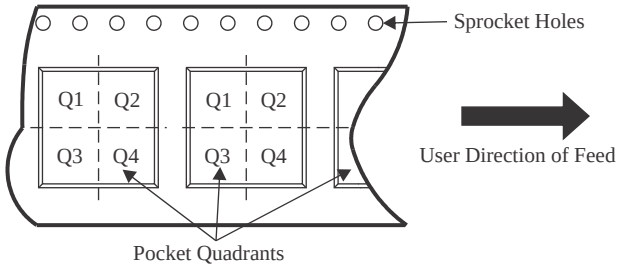


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

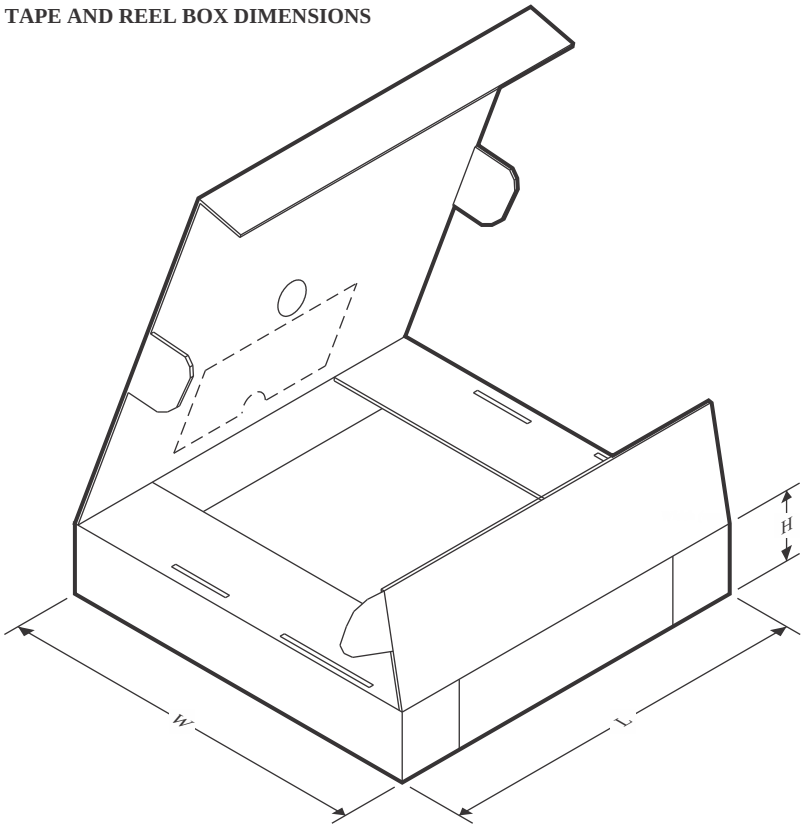
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
DAC714U/1K	SOIC	DW	16	1000	330.0	16.4	10.75	10.7	2.7	12.0	16.0	Q1

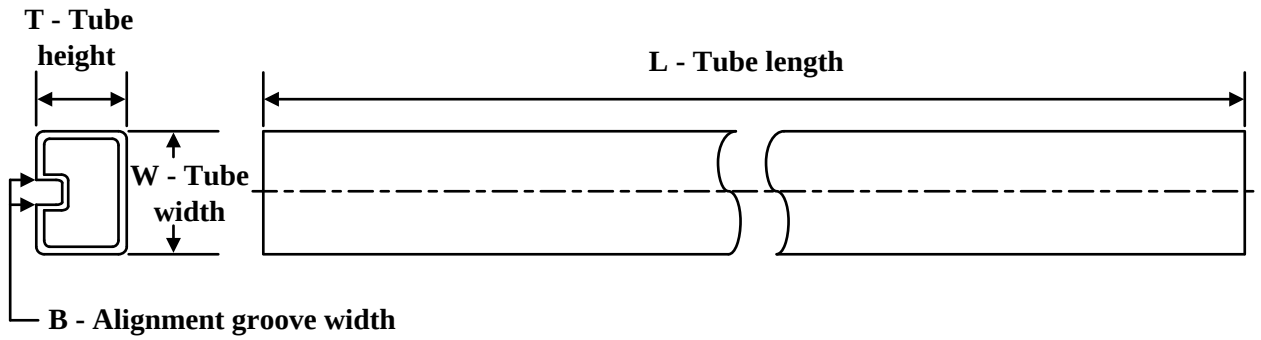
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
DAC714U/1K	SOIC	DW	16	1000	353.0	353.0	32.0

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
DAC714P	N	PDIP	16	25	506	13.97	11230	4.32
DAC714P.A	N	PDIP	16	25	506	13.97	11230	4.32
DAC714U	DW	SOIC	16	40	507	12.83	5080	6.6
DAC714U.A	DW	SOIC	16	40	507	12.83	5080	6.6

GENERIC PACKAGE VIEW

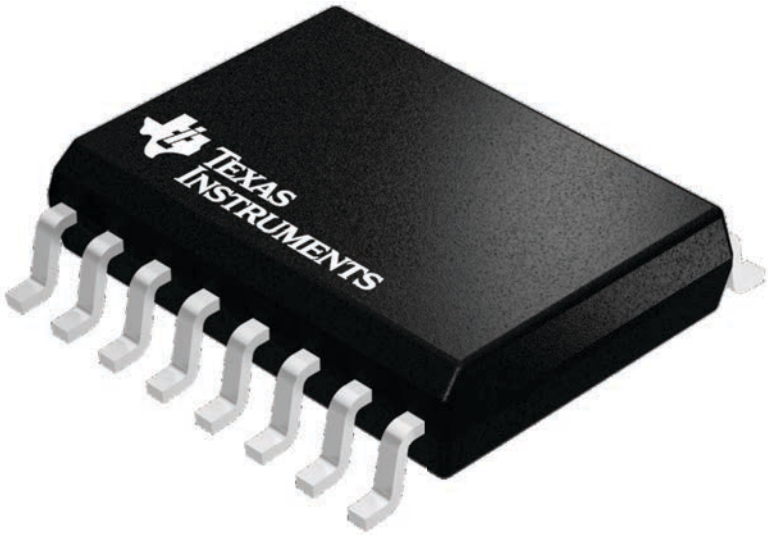
DW 16

SOIC - 2.65 mm max height

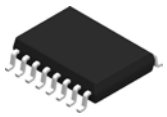
7.5 x 10.3, 1.27 mm pitch

SMALL OUTLINE INTEGRATED CIRCUIT

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



4224780/A

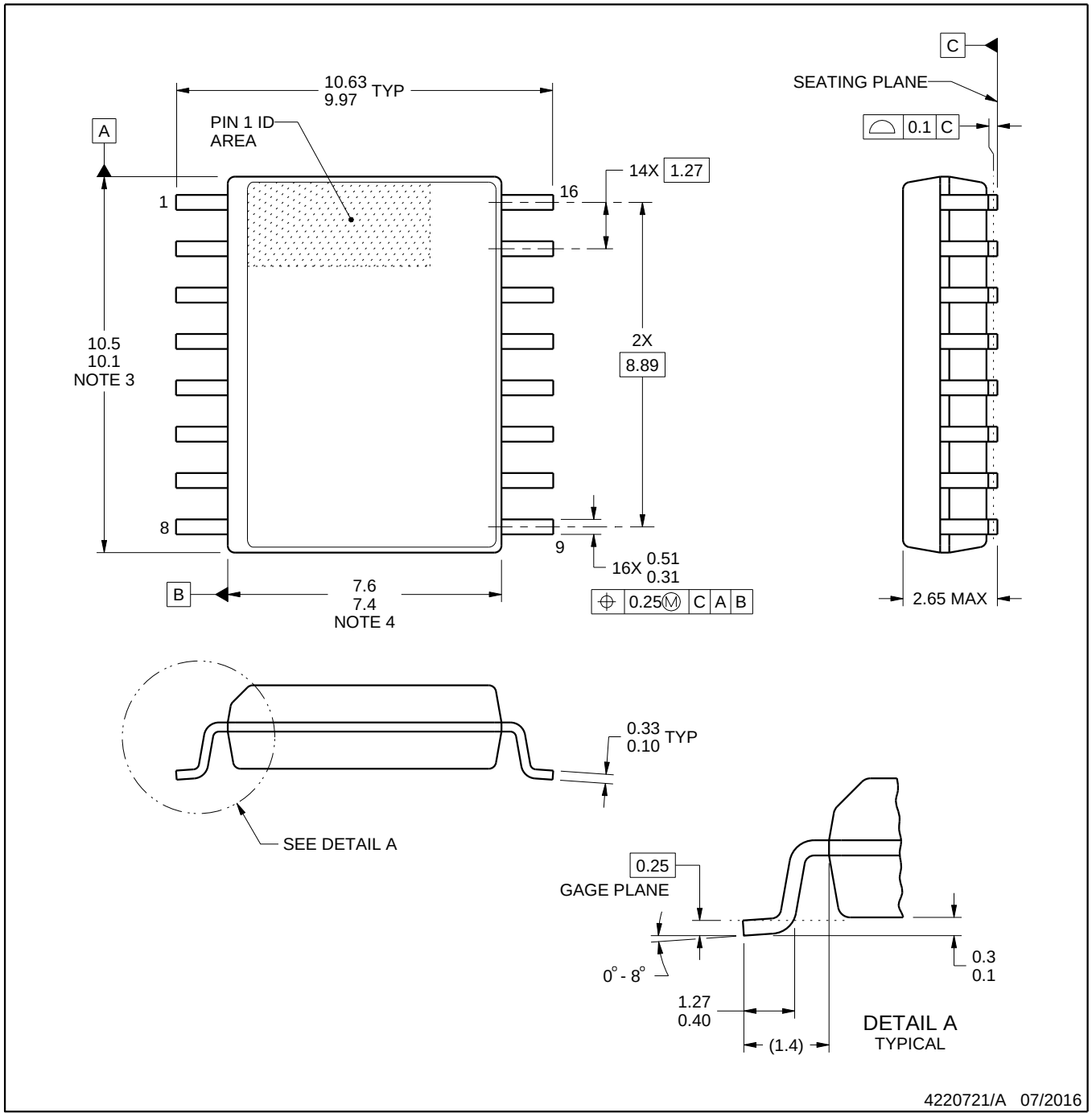


PACKAGE OUTLINE

DW0016A

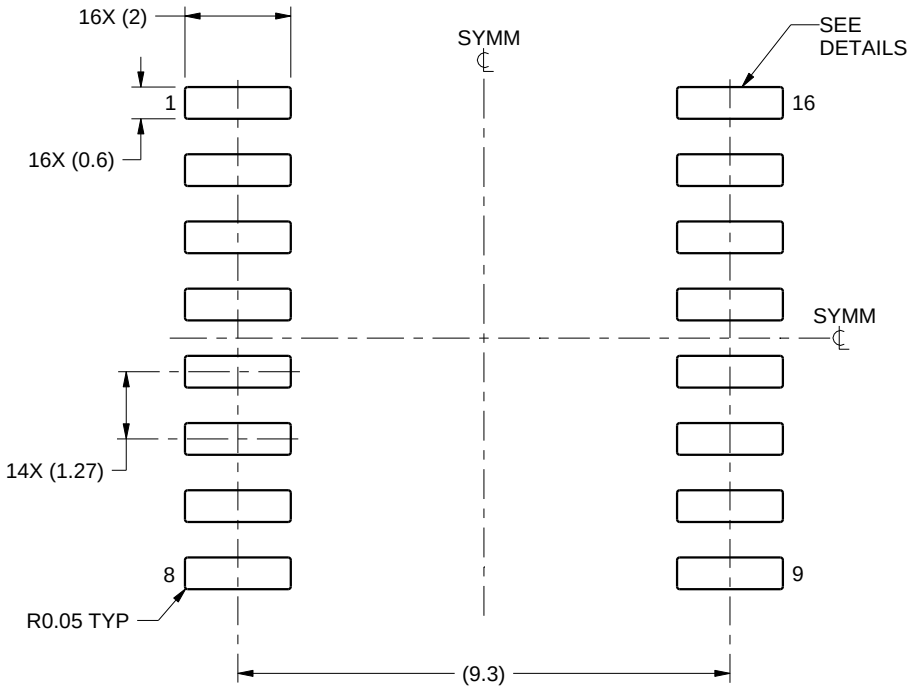
SOIC - 2.65 mm max height

SOIC

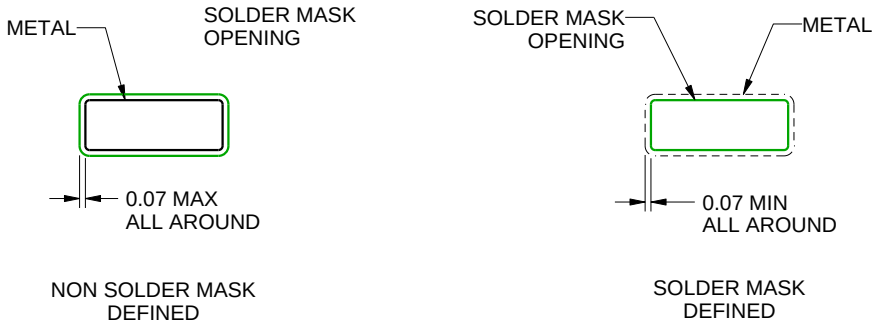


NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.
5. Reference JEDEC registration MS-013.



LAND PATTERN EXAMPLE
SCALE:7X

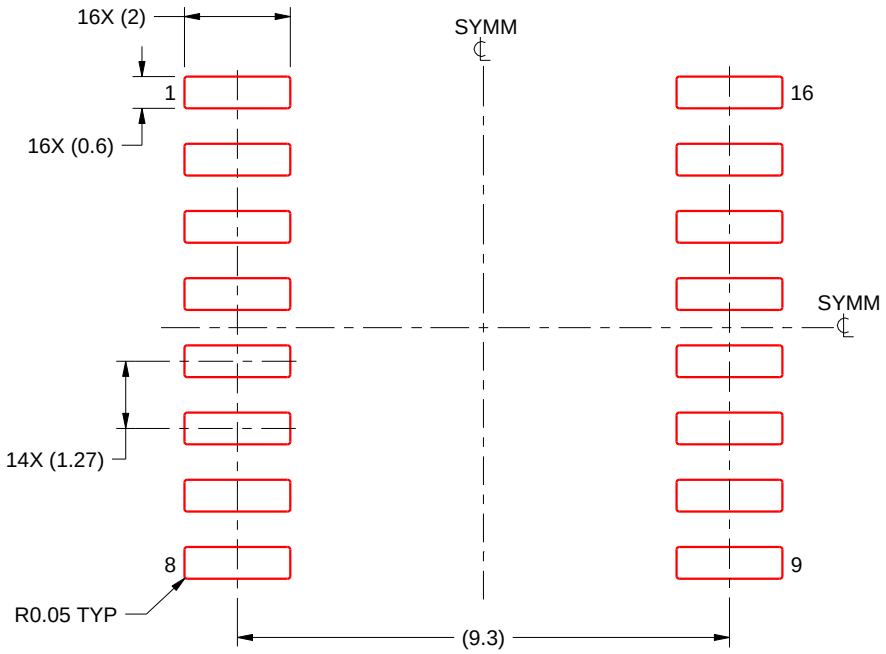


SOLDER MASK DETAILS

4220721/A 07/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:7X

4220721/A 07/2016

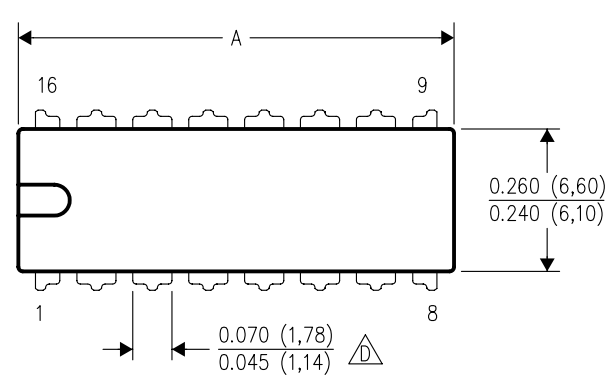
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

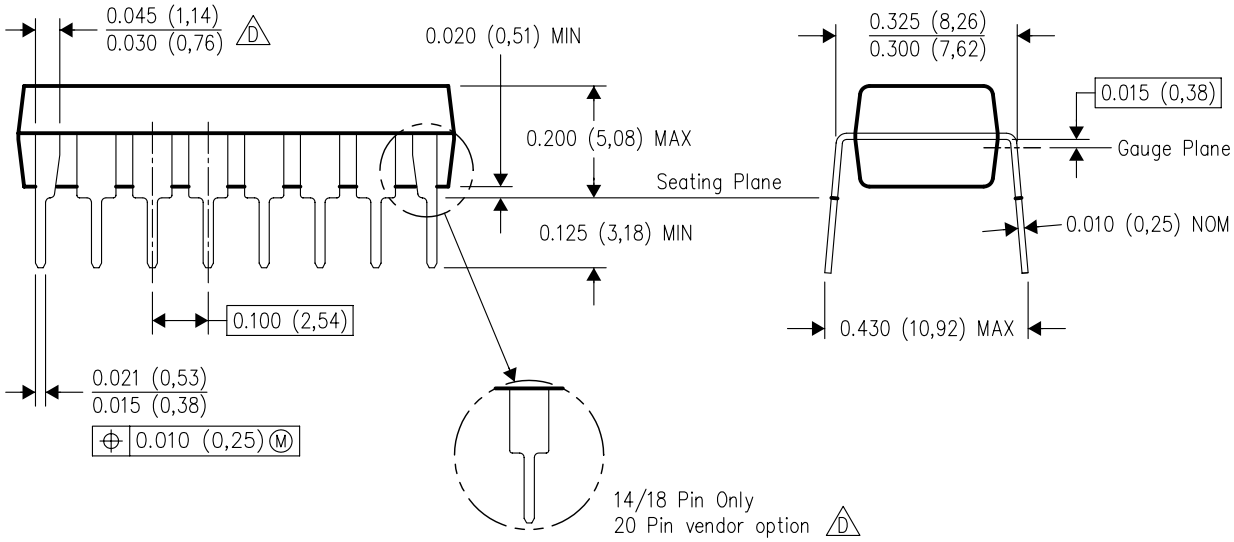
N (R–PDIP–T**)

PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN



DIM \ PINS **	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS–001 VARIATION (C)	AA	BB	AC	AD



4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

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